Lab-4 — Scenario 20: Mastermind

Vibe Coding / LLM Workflow Record

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Purpose

This record documents the incremental LLM-assisted workflow used to fix the supplied Mastermind project and add the required features while preserving the original modular structure.

Prompt 1 — Inspect and fix the original defect

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Inspect the existing Mastermind project before changing it. The current logic uses:
partial = sum(ch in code for ch in guess) - exact
Fix the duplicate-symbol feedback bug so that exact matches are resolved first and each occurrence in the c
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Prompt 2 — Add the requested features

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Using the existing corrected Mastermind project, add three difficulty modes that vary code length, symbol r
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Prompt 3 — Test and finalize

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Thoroughly test the Mastermind implementation for exact matches, repeated symbols, no matches, mixed exact/
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Implemented result

Area	Completed behavior
Feedback	Exact-first, duplicate-aware partial matching; each code occurrence used at most once.
Lifecycle	Win, loss, quit, turn limit and game-over state handled explicitly.
Difficulty	Easy, Medium and Hard with different lengths, symbol ranges and turn limits.
Input	Invalid guesses rejected without consuming turns.
History	Accepted guesses and their feedback are preserved and displayed.
Dependencies	Python standard library only; no persistence.

Testing evidence

The local test suite completed successfully. It checks exact matches, no matches, mixed exact/partial matches, duplicate-symbol cases such as code 1123 with guess 1111, all difficulty configurations, and invalid-input turn preservation.

Important submission note

This document is a prepared LLM workflow record. If the instructor specifically requires the native export of the complete Vibe Coding tool conversation, include that native export as well; this PDF should not be represented as a native export from a separate Vibe Coding platform.